Started on	Wednesday, 20 September 2023, 11:06 PM
State	Finished
Completed on	Wednesday, 20 September 2023, 11:07 PM
Time taken	54 secs
Marks	1.00/85.00
Grade	**0.75** out of 64.00 (**1**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following is the most appropriate test for short-term memory?

Select one:

- Finger tapping
- Digit span ✔
- Clock-drawing
- Facial recognition
- Verbal fluency

Your answer is correct.

EXPLANATION: Working memory is synonymous is synonymous with short-term memory, which is responsible for the immediate recall of small amounts of verbal (as in digit span) or visuospatial information.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 82.

The correct answer is: Digit span

Question 2

Not answered

Marked out of 1.00

Which type of memory is most relatively well preserved in schizophrenia?

Select one:

- Short-term
- Long-term
- Episodic
- Procedural
- Semantic

Your answer is incorrect.

EXPLANATION: Spatial working memory, short-term and long-term memory, working memory, episodic memory, and semantic memory have all been found to be impaired in people with schizophrenia, compared to health controls.

Ref: Gebreegziabhere Y et al. Cognitive impairment in people with schizophrenia: an umbrella review. European Archives of Psychiatry and Clinical Neuroscience. 2022 Oct;272(7):1139-55.

The correct answer is: Procedural

Question 3

Not answered

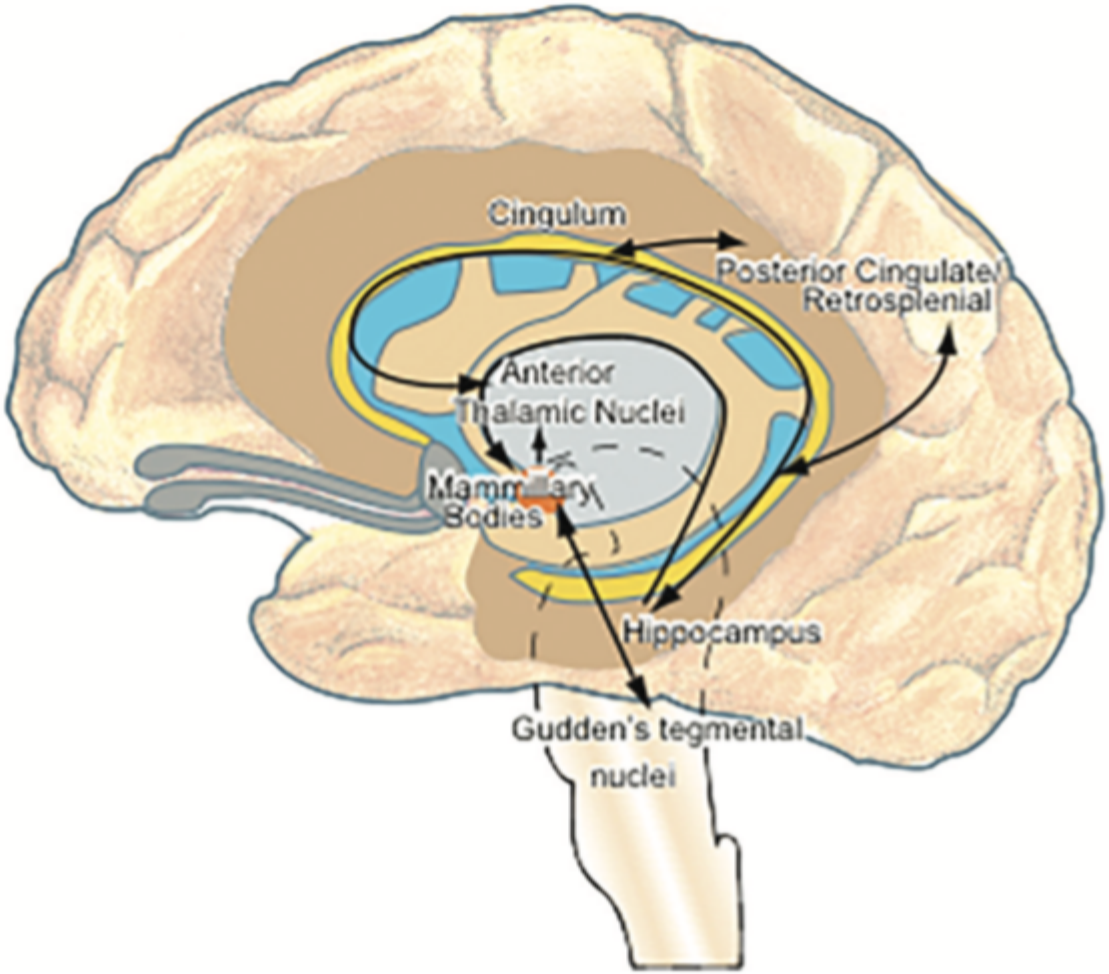
Marked out of 1.00

What is the pathophysiologic mechanism of Wernicke-Korsakoff syndrome?

Select one:

- Substantia nigra mercury poisoning
- Peripheral nervous system demyelination
- Mamillary body alpha-keto glutamate dysfunction
- Cerebellar free radical toxicity
- Neural tube homocysteine dysfunction

Your answer is incorrect.



EXPLANATION: Wernicke encephalopathy (WE) and Korsakoff psychosis, together termed Wernicke-Korsakoff syndrome, are distinct yet overlapping neuropsychiatric disorders associated with thiamine deficiency. Deficiency of thiamine can lead to decreased functioning of enzymes involved in the tricarboxylic acid cycle (alpha-keto glutamate dehydrogenase and pyruvate dehydrogenase), pentose phosphate pathway, and branched-chain amino acid metabolism. Mammillary body lesions are the classic autopsy sign in WE patients and are present in almost all cases.

Location of mammillary bodies is shown in the attached figure adapted from Meys et al. American Journal of Neuroradiology. 2022 Jun 1;43(6):802-12.

Ref: Chandrakumar A et al. Review of thiamine deficiency disorders: Wernicke encephalopathy and Korsakoff psychosis. Journal of basic and clinical physiology and pharmacology. 2018 Oct 2;30(2):153-62.

The correct answer is: Mamillary body alpha-keto glutamate dysfunction

Question **4**

Not answered

Marked out of 1.00

It is understood that the consolidation of emotional memories occurs in which stage of sleep?

Select one:

- Stage 1
- Stage 2
- Stage 3
- Consolidation of memories does not occur during sleep
- REM

Your answer is incorrect.

EXPLANATION: Evidence for the role of REM sleep in consolidation of the emotional components of memory is relatively strong. The precise mechanisms by which REM sleep promotes consolidation of emotional memories remains unresolved, although there are some studies suggesting that long-term potentiation processes are activated in REM sleep. There is a large body of literature suggesting that theta oscillations, characteristic of REM sleep, play a role in memory consolidation, particularly in fear conditioning and emotional memories.

Ref: Sara SJ. Sleep to remember. Journal of Neuroscience. 2017 Jan 18;37(3):457-63.

The correct answer is: REM

Question 5

Not answered

Marked out of 1.00

What is the pathophysiologic mechanism of multisystem atrophy?

Select one:

- Alpha-synucleinopathy
- Polyglutamate disease
- Medication-induced parkinsonism
- Hepatolenticular degeneration
- Tauopathy

Your answer is incorrect.

EXPLANATION: Multisystem atrophy (MSA), a family of Parkinson-plus diseases, includes a cerebellar type (formerly olivopontocerebellar degeneration) that is notable for ataxia and a subtype with pronounced autonomic dysfunction (formerly Shy-Drager syndrome). There is a common neuropathological finding of distinctive inclusion bodies in glial cells which contain alpha-synuclein, leading to the classification of MSA as a synucleinopathy. No mutations in the gene for alpha-synuclein have been found in MSA, which appears to exist only as a sporadic disorder.

Ref: Kaufman DM, Geyer HL, Milstein MJ, Rosengard JL. Kaufman's Clinical Neurology for Psychiatrists, 9th Edition. 2023. p. 423.

Gray F, Duyckaerts C, de Girolami U. Escourolle and Poirier's Manual of Basic Neuropathology, 6th Edition. 2019. p. 204.

The correct answer is: Alpha-synucleinopathy

Question 6

Not answered

Marked out of 1.00

Which of the following is an important component of the blood brain barrier (BBB)?

Select one:

- APP
- Huntingtin
- Hypocretin
- ApoE
- Prion protein

Your answer is incorrect.

EXPLANATION: Cerebrovascular homeostasis is maintained by the blood-brain barrier (BBB), a highly selective structure that separates the peripheral blood circulation from the brain and protects the central nervous system. Several recent studies have highlighted the importance of the physiological role of the beta-amyloid precursor protein (APP) in endothelial cell homeostasis, suggesting an important role of this protein in maintaining vascular stability.

Ref: Ristori E et al. New insights into blood-brain barrier maintenance: the homeostatic role of ß-amyloid precursor protein in cerebral vasculature. Frontiers in Physiology. 2020 Aug 27;11:1056.

The correct answer is: APP

Question 7

Not answered

Marked out of 1.00

Brain organoids can mimic which age of the human brain?

Select one:

- 2nd trimester
- 10 years old
- 20 years old
- 6 days old
- 1 year old

Your answer is incorrect.

EXPLANATION: Brain organoids are produced by the differentiation of pluripotent stem cells under three-dimensional culture conditions by adding neurodevelopment- related regulatory signals. They are similar to the cell composition and anatomical structure of the brain, and can reflect the developmental process of the bring, as well as their physiology, pathology, and pharmacology. In terms of their cellular and molecular composition, the current architecture of brain organoids can mimic the second-trimester human fetal brain.

Ref: Chen H et al. Brain organoids: Establishment and application. Frontiers in Cell and Developmental Biology. 2022 Nov 23;10:1029873.

The correct answer is: 2nd trimester

Question 8

Not answered

Marked out of 1.00

Dysfunction in which neurotransmitter is likely to contribute the most to cognitive symptoms in schizophrenia?

Select one:

- Acetylcholine
- Dopamine
- Serotonin
- Glutamate
- Norepinephrine

Your answer is incorrect.

EXPLANATION: Although dopamine dysfunction has mainly been linked to positive psychotic symptoms, striatal dopamine signalling normally has widespread effects on cortical function, and its dysregulation can also cause cognitive symptoms. Many researchers believe that cognitive and some negative symptoms of schizophrenia may be due to a deficit of dopamine activity in mesocortical projections to the dorsolateral prefrontal cortex.

Ref: McCutcheon RA et al. Cognitive impairment in schizophrenia: aetiology, pathophysiology, and treatment. Molecular psychiatry. 2023 Jan 23:1-7.

Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 95.

The correct answer is: Dopamine

Question 9

Not answered

Marked out of 1.00

The processing of empathy is most influenced by which of the following structures?

Select one:

- Hypothalamus
- Cerebellum
- Insula
- DLPFC
- Amygdala

Your answer is incorrect.

EXPLANATION: The anterior insula cortex has been shown to be a core circuit for empathy.

Ref: Li Y et al. Linking brain structure and activation in anterior insula cortex to explain the trait empathy for pain. Human brain mapping. 2020 Mar;41(4):1030-42.

The correct answer is: Insula

Question 10

Not answered

Marked out of 1.00

Which of the following is recommended for the treatment of REM sleep behaviour disorder?

Select one:

- Amitriptyline
- Deep brain stimulation
- Melatonin
- Prazosin
- Trazodone

Your answer is incorrect.

EXPLANATION: Recommended treatments for REM sleep behaviour disorder include clonazepam, melatonin, and pramipexole. Melatonin binds to the M1 and M2 receptors, suppressing REM sleep motor tone and renormalising other circadian features of REM sleep. Melatonin requires a prescription in the EU and the UK.

Ref: Howell M. Management of REM sleep behavior disorder: an American Academy of Sleep Medicine clinical practice guideline. Journal of Clinical Sleep Medicine. 2023 Apr 1;19(4):759-68.

The correct answer is: Melatonin

Question 11

Not answered

Marked out of 1.00

Loss of orexin-producing neurons takes place in which condition?

Select one:

- Amyotrophic lateral sclerosis
- Wilson disease
- Narcolepsy
- Parkinson disease
- Huntington disease

Your answer is incorrect.

EXPLANATION: Orexin/hypocretin neurons are localised exclusively in certain hypothalamic areas (later hypothalamic area, perifornical area, and posterior hypothalamus). These hypothalamic neurons degenerate in narcolepsy, and loss of these neurons causes the inability of orexin to be produced and released downstream on wake-promoting neurotransmitter centres and thus lack of stabilising wakefulness.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 407.

The correct answer is: Narcolepsy

Question 12

Not answered

Marked out of 1.00

In Bayesian theory, what is the term given to an expected result or belief?

Select one:

- Data likelihood
- Posteriors
- Prediction error
- Priors
- Anteriors

Your answer is incorrect.

EXPLANATION: The account of predictive processing has emerged as an alluring computational model to explain how brain processes information about the world. In the higher brain centers, expected predictions and pre-existing beliefs (known as priors) are said to be combined with observed sensory data (referred to as likelihood) to compute a posterior probability (referred to as the posterior). This posterior probability is said to represent the most probable perception, considering both the prior beliefs and the sensory evidence. In cases where there is a disparity between the prior expectations and the sensory evidence (likelihood), a prediction error is generated. This error signal initiates the necessity to revise the brain's current model and adjust its prior beliefs accordingly. To express it formally, a prediction error quantifies the difference between the means of the prior and likelihood distributions, taking into account their respective precisions (inverse variance).

Ref: Wilson P, Humpston C, Nathan R. Innovations in the psychopathology of schizophrenia: a primer for busy clinicians. BJPsych Advances. 2022 Mar;28(2):79-89.

The correct answer is: Priors

Question 13

Not answered

Marked out of 1.00

Which of the following protein inclusion profiles is most common in FTD?

Select one:

- Tau positive, amyloid positive
- Synuclein positive, tau positive
- Ubiquitin negative, tau negative
- Ubiquitin positive, tau positive
- Tau positive, amyloid negative

Your answer is incorrect.

EXPLANATION: The classification of frontotemporal dementias is based on the type of protein inclusions observed in neurons as well as the mutational status, if known. The inclusions contain predominantly or exclusively one of three markers: tau; TDP-43, a DNA/RNA-binding protein; and FUS (fused in sarcoma). Atypical frontotemporal lobar degeneration with ubiquitin-positive inclusion is a relatively rare cause of FTD characterised by FUS-positive inclusions.

Ref: Gray F, Duyckaerts C, de Girolami U. Escourolle and Poirier's Manual of Basic Neuropathology, 6th Edition. 2019. p. 192. Peet BT et al. Neuroimaging in frontotemporal dementia: heterogeneity and relationships with underlying neuropathology. Neurotherapeutics. 2021 Apr;18:728-52.

The correct answer is: Tau positive, amyloid negative

Question 14

Not answered

Marked out of 1.00

How many times does a G protein coupled receptor pass through the cell membrane?

Select one:

- 7 times
- 2 times
- 4 times
- 5 times
- 12 times

Your answer is incorrect.

EXPLANATION:Receptors linked to G proteins all have the structure of seven-transmembrane regions, meaning that they span the membrane seven times.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 36.

The correct answer is: 7 times

Question 15

Not answered

Marked out of 1.00

Purines

Select one:

- Are nitrogen-containing compounds
- Include cytosine and thymine
- Are found only in eukaryotes
- Can only be obtained from the diet
- Include ethanol and nicotine

Your answer is incorrect.

EXPLANATION: A purine is an aromatic heterocycle composed of carbon and nitrogen. Eukaryotes, bacteria, and archaea are all able to carry out de-novo biosynthesis of purines. Purines include adenine and guanine, which participate in DNA and RNA formation. Purines are also constituents of other important biomolecules, such as ATP, GTP, cyclic AMP, NADH, and coenzyme A. Other important purines include hypoxanthine, xanthine, theobromine, caffeine, uric acid, and isoguanine. Cytosine, thymine, and uracil are pyrimidines.

Ref: Kumari, A. Chapter 17 - Purine Structures in Sweet Biochemistry, 2018. pp. 89-91.

The correct answer is: Are nitrogen-containing compounds

Question 16

Not answered

Marked out of 1.00

Which of the following candidate genes form the most biologically plausible genetic basis of schizophrenia?

Select one:

- DTNBP1 and GAD67
- DISC1 and NRG1
- GLO1 and ZNF804A
- ZHHC8 and AKT1
- DAOA and LIS1

Your answer is incorrect.

EXPLANATION: DISC1 is one of the most important candidate genes for schizophrenia. It is one of the most researched genes in schizophrenia, alongside the COMT AND NRG1 genes. DISC1 is a protein that performs many functions, including neuronal migration, intracellular transport, synaptic maintenance, and stabilisation. NRG1 is one of the most important proteins for the variety of functions in which it is involved. Among them are the growth and development of the epidermis and development of the nervous system (synaptic plasticity).

Ref: Nastase MG et al. Genetic polymorphism and neuroanatomical changes in schizophrenia. Romanian Journal of Morphology and Embryology. 2022 Apr;63(2):307.

The correct answer is: DISC1 and NRG1

Question 17

Not answered

Marked out of 1.00

Which of the following neurotransmitters has been shown to be sensitive to epigenetic tuning in the context of maternal attachment?

Select one:

- Dopamine
- Glutamate
- Norepinephrine
- Oxytocin
- Serotonin

Your answer is incorrect.

EXPLANATION: Oxytocin is a crucial regulator of human social behaviour. Individual differences within the oxytocin system – assessed through hormonal levels, epigenetic modification, or genetic variation - have been linked to differential sensitivity to social cues, prosocial behaviours, and stress responsiveness in adults. Early experience in maternal care predicts epigenetic change within the infant oxytocin system, levels of which are in turn reflective of infant temperament.

Ref: Krol KM et al. Epigenetic dynamics in infancy and the impact of maternal engagement. Science Advances. 2019 Oct 16;5(10):eaay0680.

The correct answer is: Oxytocin

Question 18

Not answered

Marked out of 1.00

Motor speed is best assessed by which of the following tests?

Select one:

- Clock drawing
- Finger tapping
- Digit span
- Trail making
- Token test

Your answer is incorrect.

EXPLANATION: Finger tapping is a standard measure of simple motor speed. It is particularly useful for documenting lateralised motor impairment.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 53.

The correct answer is: Finger tapping

Question 19

Not answered

Marked out of 1.00

A non-dominant cerebral hemisphere lesion is most likely to lead to which of the following?

Select one:

- Finger agnosia
- Constructional apraxia (Gestalt)
- Constructional apraxia (details)
- Right-left disorientation
- Limb apraxia

Your answer is incorrect.

EXPLANATION: Symptoms arising from non-dominant hemisphere injury tend to be subtle, and predominantly neuropsychologic. The cause is usually a structural lesion, such as trauma, stroke, or malignant tumour, that has rapidly developed in the non-dominant parietal or frontal lobe cortex. Neuropsychological deficits include constructional apraxia (Gestalt), dressing apraxia, and anosognosia.

Ref: Kaufman DM, Geyer HL, Milstein MJ, Rosengard JL. Kaufman's Clinical Neurology for Psychiatrists, 9th Edition. 2023. pp. 168-169.

The correct answer is: Constructional apraxia (Gestalt)

Question 20

Not answered

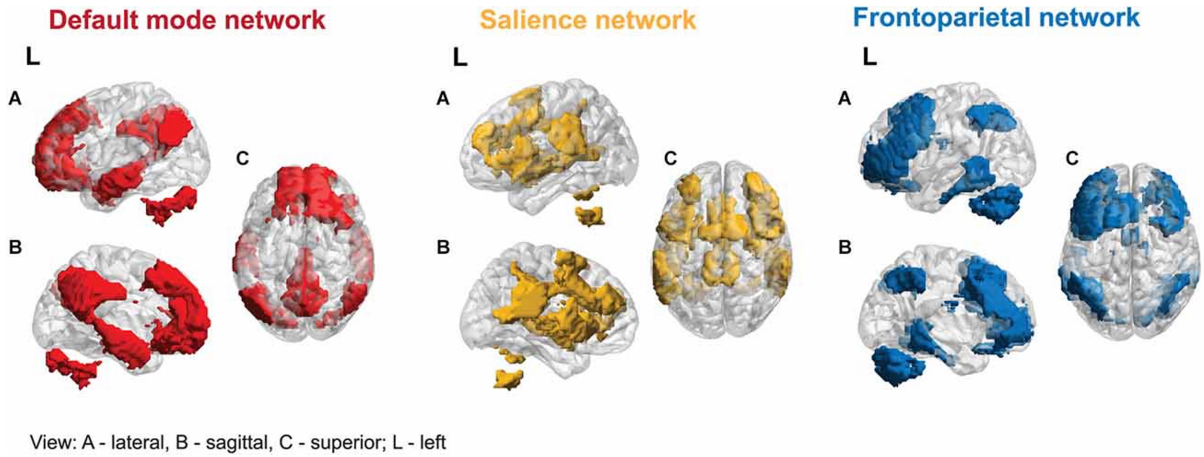
Marked out of 1.00

Which of the following is part of the default mode network?

Select one:

- Globus pallidus
- Inferior parietal lobe
- Occipital lobe
- Nucleus accumbens
- Caudate nucleus

Your answer is incorrect.



View: A - lateral, B - sagittal, C - superior; L - left

EXPLANATION: The default mode network is an integrated system of cortical areas that are active in a person who is in a state of wakeful rest, daydreaming, reminiscing, or contemplating future events, but not focused on a specific task involving the sensory (especially visual) input. It includes the medial prefrontal cortex, the posterior cingulate gyrus, and angular gyrus, and the medial, lateral, and inferior parietal lobe (IPL), and parts of the parahippocampal gyrus.

The attached picture (from Schimmelpfennig J, et al. Frontiers in human neuroscience. 2023 Mar 20;17:1133367) displays the other major neurocognitive networks; note that IPL is not exclusively a DMN-only region.

Ref: Menon V. 20 years of the default mode network: A review and synthesis. Neuron. 2023 May 10.

The correct answer is: Inferior parietal lobe

Question 21

Not answered

Marked out of 1.00

Which of the following is the best test for selective attention?

Select one:

- Story recall
- Rey-Osterrieth complex figure test
- Raven's progressive matrices
- National adult reading test
- Stroop test

Your answer is incorrect.

EXPLANATION: Stroop tests are based on the fact that it takes longer to call out the colour names of coloured patches than to read words, and even longer to name the colour of an ink in which a colour name is printed when the print ink is a colour different from the colour name. The latter phenomenon, a markedly slowed response when a colour name is printed in ink of a different colour, is interpreted as a failure of selective attention.

Ref: Hodges JR. Cognitive assessment for clinicians, 3rd Edition. 2018. p. 248.

The correct answer is: Stroop test

Question 22

Not answered

Marked out of 1.00

According to ICD-11, how many episodes of depression are needed for a diagnosis of recurrent depressive disorder?

Select one:

- One
- Three
- Four
- Five
- Two

Your answer is incorrect.

EXPLANATION: In ICD-11, recurrent depressive disorder is characterised by a history of at least two depressive episodes separated by at least several months without significant mood disturbance.

Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed. ICD-11; World Health Organization, 2020)

The correct answer is: Two

Question 23

Not answered

Marked out of 1.00

In ICD-11, which of the following features is characteristic of children with reactive attachment disorder?

Select one:

- Cognitive deficits
- Social communication deficits
- Minimal seeking of comfort
- Fearfulness
- Hyperactivity and inattention

Your answer is incorrect.

EXPLANATION: In ICD-11, reactive attachment disorder is characterised by grossly abnormal behaviours in early childhood, occurring in the context of a history of grossly inadequate child care. Even when an adequate primary caregiver is newly available, the child does not turn to the primary caregiver for comfort, support, and nurture, rarely displays security-seeking behaviours towards any adult, and does not respond when comfort is offered.

Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed. ICD-11; World Health Organization, 2020)

The correct answer is: Minimal seeking of comfort

Question 24

Not answered

Marked out of 1.00

According to ICD-11, which of the following may be seen in OCD but not in hoarding disorder?

Select one:

- Pleasure and enjoyment in hoarding
- Hoarding motivated by obsessional thoughts
- Accumulation of excessive amounts of objects
- Hoarded objects have an intrinsic value
- Compulsive hoarding

Your answer is incorrect.

EXPLANATION: Individuals with OCD may accumulate excessive amounts of objects (i.e. compulsive hoarding). However, unlike hoarding disorder, the behaviour is undertaken with the goal of neutralising or reducing concomitant distress and anxiety arising from obsessional content such as aggressive, sexual/religious, contamination, or symmetry/ordering themes. Furthermore, even in individuals affected by OCD who have poor or absent insight, the behaviour is generally unwanted and distressing, whereas in hoarding disorder it may be associated with pleasure or enjoyment. Generally in hoarding disorder, items are hoarded because of their emotional significance, instrumental characteristics, or intrinsic value (e.g. perceived aesthetic qualities).

Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed. ICD-11; World Health Organization, 2020)

The correct answer is: Hoarding motivated by obsessional thoughts

Question 25

Not answered

Marked out of 1.00

In ICD-11, what is the minimum duration of grief response required for a diagnosis of prolonged grief disorder to be made?

Select one:

- >6 months
- >1 month
- >1 year
- >2 years
- >5 years

Your answer is incorrect.

EXPLANATION: Prolonged grief disorder is a disturbance in which, following the death of a person close to the bereaved, there is a persistent and pervasive grief response characterised by longing for the deceased or persistent preoccupation with the deceased accompanied by intense emotional pain. In ICD-11, the grief response must have persisted for an atypically long period of time following the loss (more than 6 months at a minimum).

Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed. ICD-11; World Health Organization, 2020)

The correct answer is: >6 months

Question 26

Not answered

Marked out of 1.00

Which of the following is a specifier for a depressive episode in ICD-11?

Select one:

- With pseudodementia
- With biological symptoms
- With suicidality
- With grief
- With melancholia

Your answer is incorrect.

EXPLANATION: In ICD-11, available specifiers for a depressive episode include: with prominent anxiety symptoms; with panic attacks; persistent; with melancholia; with seasonal pattern.

Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed. ICD-11; World Health Organization, 2020)

The correct answer is: With melancholia

Question 27

Not answered

Marked out of 1.00

As defined in ICD-11, which of the following best differentiates complex PTSD from PTSD?

Select one:

- Avoidance of memories of traumatic events
- Re-experiencing of traumatic events
- Dysphoria
- Persistent difficulties sustaining relationships
- A series of continued traumatic events

Your answer is incorrect.

EXPLANATION: Whereas the ICD-11 diagnostic requirements for complex PTSD include all essential features of PTSD, the diagnosis of complex PTSD also requires the additional essential features of severe problems in affect regulation, persistent negative beliefs about oneself, and persistent difficulties in sustaining relationships.

Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed. ICD-11; World Health Organization, 2020)

The correct answer is: Persistent difficulties sustaining relationships

Question 28

Not answered

Marked out of 1.00

Which of the following is an interviewer-rated scale?

Select one:

- Geriatric Depression Scale
- HAM-D
- HADS
- General Health Questionnaire
- PHQ-9

Your answer is incorrect.

EXPLANATION: The Hamilton Rating Scale for Depression (HAM-D) is an interviewer-rated, 17-item rating scale for depressive illness. It is not a diagnostic instrument, but is used to measure changes (e.g. as a result of drug treatment). 17 items are scored according to severity, producing the final score.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 97.

The correct answer is: HAM-D

Question 29

Not answered

Marked out of 1.00

What is an advantage of using a cross-sectional method to study developmental phenomena in children?

Select one:

- High response rates
- Ability to infer causality
- Avoids the confounding effect of age
- Quick data collection
- Longitudinal insights

Your answer is incorrect.

EXPLANATION: One advantage of using a cross-sectional method in studying developmental phenomena in children is that it allows for quick data collection from participants of different age groups at a single point in time. This method is efficient for obtaining a snapshot of developmental trends across different age cohorts, but it does not provide longitudinal insights into individual development or establish temporal causality. Attrition rates can vary, and the use of this design alone does not inherently improve response rates.

The correct answer is: Quick data collection

Question 30

Not answered

Marked out of 1.00

Which of the following is a test of pre-morbid intelligence?

Select one:

- CANTAB Battery
- Autobiographical Memory Interview
- National Adult Reading Test
- Behavioural Inattention Test
- Boston Naming Test

Your answer is incorrect.

EXPLANATION: Since the ability to pronounce irregular words correctly is highly educationally dependent, a test of irregular word reading, the National Adult Reading Test (NART), is widely used to predict premorbid intellectual disability.

Ref: Hodges JR. Cognitive assessment for clinicians, 3rd Edition. 2018. p. 80.

The correct answer is: National Adult Reading Test

Question 31

Not answered

Marked out of 1.00

Which of the following is the most appropriate test for abstract reasoning?

Select one:

- Serial 7s test
- Digit span
- Trail-making test
- Proverb testing
- Cognitive estimates

Your answer is incorrect.

EXPLANATION: Abstract reasoning is the ability to shift back and forth between general concepts and specific examples. Having the patient identify similarities between like objects or concepts (apple and pear, bus and airplane, or a poem and a painting) as well as interpreting proverbs are useful ways to assess one's ability to abstract.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 11.

The correct answer is: Proverb testing

Question 32

Not answered

Marked out of 1.00

According to ICD-11, which of the following is seen in anorexia nervosa, but not in avoidant-restrictive food intake disorder?

Select one:

- Significantly low body weight
- Significant functional impairment
- Health-related consequences
- Restricted eating
- Weight loss to achieve thinness

Your answer is incorrect.

EXPLANATION: Individuals with anorexia nervosa (AN), like individuals with avoidant-restrictive food intake disorder (ARFID), present with a pattern of restricted eating and significantly low body weight, with similar health-related consequences. The difference is that in AN, behaviours to establish or maintain an abnormally low body weight are usually explicitly motivated by a desire for thinness or an intense fear of gaining weight. In ARFID, the pattern of eating behaviour is not motivated by preoccupation with body weight or shape.

Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed. ICD-11; World Health Organization, 2020)

The correct answer is: Weight loss to achieve thinness

Question 33

Not answered

Marked out of 1.00

Which of the following statements best describes the concept of group polarisation?

Select one:

- The tendency of a group to select individuals to blame for adverse outcomes
- The tendency for a group to make a more extreme decision than its individual members would have done initially
- Individuals losing self-awareness while in a group setting
- The tendency of a group to favour its own members over others
- The tendency for a group to engage in irrational decision-making to preserve harmony

Your answer is incorrect.

EXPLANATION: Group polarisation is defined within social psychology as the tendency for groups to move towards a more extreme version of their initial beliefs following group discussion, e.g. being more risky if they were originally inclined to be, and more cautious if this was their initial inclination.

Ref: Arvan, M. 2019. 'The Dark Side of Morality: Group Polarization and Moral Epistemology'. Philosophical Forum 50 :87- 115.

The correct answer is: The tendency for a group to make a more extreme decision than its individual members would have done initially

Question 34

Not answered

Marked out of 1.00

A young child appears frozen after hearing a loud noise but does not come close when offered comfort. What is the most likely diagnosis?

Select one:

- OCD
- Severe intellectual disability
- Depressive disorder
- Social anxiety disorder
- Reactive attachment disorder

Your answer is incorrect.

EXPLANATION: In ICD-11, reactive attachment disorder is characterised by grossly abnormal behaviours in early childhood, occurring in the context of a history of grossly inadequate child care. Even when an adequate primary caregiver is newly available, the child does not turn to the primary caregiver for comfort, support, and nurture, rarely displays security-seeking behaviours towards any adult, and does not respond when comfort is offered.

Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed. ICD-11; World Health Organization, 2020)

The correct answer is: Reactive attachment disorder

Question 35

Not answered

Marked out of 1.00

When public health campaigns aim to influence behaviour by detailing the harms of smoking and the benefits of not smoking, this is best attributed to which type of power, according to classic work of French & Raven?

Select one:

- Referent
- Informational
- Expert
- Coercive
- Legitimate

Your answer is incorrect.

EXPLANATION: According to French & Raven, informational power stems from the logical or persuasive content of the communication between the influencing agent and the target. Informational power is used when campaigns explain the reasons for recommended behaviours, indicating how such behaviours would contribute to improved heath.

Ref: Brown JH, Raven BH. Power and Compliance in Doctor/Patients Relationship. REVISTA LEX MERCATORIA Doctrina, Praxis, Jurisprudencia y Legislación. 1994;6(1):3-22.

The correct answer is: Informational

Question 36

Not answered

Marked out of 1.00

The unintentional distortion of memory that occurs when it is filtered through a person's current emotional, experiential, and cognitive state is best referred to as

Select one:

- False memory
- Retrospective delusion
- Pseudologia fantastica
- Screen memory
- Retrospective falsification

Your answer is incorrect.

EXPLANATION: Retrospective falsification refers to the unintentional distortion of memory that occurs when it is filtered through a person's current emotional, experiential, and cognitive state. It is often found in those with depressive illness who describe all past experiences in negative terms due to the impact of their current mood. Any psychiatric illness can lead to retrospective falsification, which is related to insight of the patient as well as to suggestibility.

Ref: Casey P, Kelly B. Fish's clinical pathology: Signs and symptoms in psychiatry, 4th Edition. 2019. p. 66.

The correct answer is: Retrospective falsification

Question 37

Not answered

Marked out of 1.00

What Gestalt principle of perception is shown in this figure?



Select one:

- Closure
- Similarity
- Continuation
- Proximity
- Good figure

Your answer is incorrect.

EXPLANATION: Proximity is one of the four original grouping laws of Gestalt psychology that explain the organisation of parts into whole by the visual system. According to the law, elements close together tend to be grouped perceptually, so that the array OO OO OO OO tends to be perceived as four pairs rather than eight separate elements.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015. p. 611.

The correct answer is: Proximity

Question 38

Not answered

Marked out of 1.00

Hypoplasia of which of the following structures has been associated with autism spectrum disorder?

Select one:

- Temporal lobe
- Cerebellum
- Occipital lobe
- Hypothalamus
- Thalamus

Your answer is incorrect.

EXPLANATION: Cerebellar hypoplasia was an early documented change in brain morphology in ASD (autism spectrum disorder). The finding of cerebellar hypoplasia in ASD has been replicated and has been associated with the loss of cerebellar Purkinje cells in the posterior lobe whose degenerated myelinated axons contribute to reduced cerebellar volume.

Ref: Welsh JP et al. Intellectual disability and cerebellar hypoplasia in autism reported by associative learning. medRxiv. 2021 Mar 28:2021-03.

The correct answer is: Cerebellum

Question 39

Not answered

Marked out of 1.00

The cocktail party phenomenon involves

Select one:

- Cognitive misattribution
- Involuntary attention
- Visual perception
- Gustatory perception
- Selective auditory attention

Your answer is incorrect.

The cocktail party phenomenon involves selective attention in speech perception that enables a listener to attend to one among several equally loud conversations occurring simultaneously, factors such as voice quality and directional cues facilitating the task. Recordings of conversations under similar circumstance are very much harder to understand. The term was introduced in 1957 by the British telecommunications engineer Colin Cherry.Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015. p. 140.

The correct answer is: Selective auditory attention

Question 40

Not answered

Marked out of 1.00

A child goes to daycare for the first time. He shows no reaction when he is left by his parents and no reaction when they come back. What type of attachment does this describe?

Select one:

- Anxious-avoidant
- Reactive
- Secure
- Disorganised
- Anxious-ambivalent

Your answer is incorrect.

EXPLANATION: Mary Ainsworth devised the Strange Situation procedure with 12- to 18-month infants, which categorised infant-parent relationships. The procedure involves separations and reunions with caregivers and observes response. In insecure (anxious)-avoidant attachment, the child seems unconcerned at separation or reunion. It is associated with unresponsive parenting.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 658-659.

The correct answer is: Anxious-avoidant

Question 41

Not answered

Marked out of 1.00

Which of the following is not a strength of the nomothetic personality theory approach (which categorises specific and fixed domains to personality clusters)?

Select one:

- Personality seen in terms of attributes
- Emphasis on individual variability
- Reliability within cultures
- Universalizability in assessment
- Reliability over time

Your answer is incorrect.

EXPLANATION: There are two main approaches to conceptualise and describe variations in personality: nomothetic and ideographic. In nomothetic approaches, personality is seen in terms of attributes shared by individuals. There are two subdivisions: type (or categorical) approaches – in which there are discrete categories of personality - and trait (or dimensional) approaches – in which a limited number of qualities, or traits, account for personality variation. Unlike nomothetic approaches, ideographic approaches on the other hand, emphasise individuality and seek to understand an individual's personality by understanding that specific individual and their development, rather than by reference to common factors. Examples include psychoanalytic, humanistic, and cognitive-behavioural approaches.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 520.

The correct answer is: Emphasis on individual variability

Question 42

Not answered

Marked out of 1.00

Which of the following is true regarding the classical categorical classification of disorders?

Select one:

- Disorders are overlapping and not distinct
- Disorders appear in more than once in the system
- It facilitates collection of epidemiological data
- It must be hierarchy-based
- It allows for varying levels of severity

Your answer is incorrect.

EXPLANATION: In classical categorical systems, each entity or disorder is distinct and appears only once in a clearly defined place. With a few exceptions, ICD and DSM are such classical categorical systems. WHO member states are responsible for reporting health statistics based on ICD to WHO (e.g. causes of morbidity and mortality, prevalence). ICD remains structured as a categorical taxonomic system because this format is necessary for its application as the classification system for global health statistics. Wide use of non-standard categories would not generate valid data for global health statistics.

Ref: Clark LA et al. Three approaches to understanding and classifying mental disorder: ICD-11, DSM-5, and the National Institute of Mental Health's Research Domain Criteria (RDoC). Psychological Science in the Public Interest. 2017 Nov;18(2):72-145.

The correct answer is: It facilitates collection of epidemiological data

Question 43

Not answered

Marked out of 1.00

Which of the following is true of the Kaiser CDC Adverse Childhood Experiences Study?

Select one:

- Parental divorce was included as an adverse childhood experience
- Psychological abuse was not included in the original study
- The study utilised a questionnaire mailed to households in the UK
- Exposure to childhood abuse and household dysfunction showed no relationship to leading causes of adult deaths
- There was a significant relationship between the number of childhood abuse categories and depression as an adult

Your answer is incorrect.

EXPLANATION: The CDC-Kaiser Permanente adverse childhood experiences (ACE) study is one of the largest studies of the long-term relationship of childhood experiences to important medical and public health problems. The ACE study questionnaire was mailed to over 13 000 adults in California, USA, with a 70% response rate. Seven categories of adverse childhood experiences were studied: psychological, physical, or sexual abuse; violence against mother; and living with household members who were substance abusers, mentally ill or suicidal, or ever imprisoned. Persons who had experienced four or more categories of childhood exposure, compared to those who had experienced none, had 4 to 12-fold increased health risks for alcoholism, drug abuse, depression, and suicide attempts. The study found a strong graded relationship between the breadth of exposure to abuse or household dysfunction during childhood and multiple risk factors for several of the leading causes of death in adults.

Ref: Felitti VJ et al. Relationship of childhood abuse and household dysfunction to many of the leading causes of death in adults: The Adverse Childhood Experiences (ACE) Study. American journal of preventive medicine. 1998 May 1;14(4):245-58.

The correct answer is: There was a significant relationship between the number of childhood abuse categories and depression as an adult

Question 44

Not answered

Marked out of 1.00

In altruism research, which of the following is a replicated finding?

Select one:

- Altruism can reduce the physical perception of pain
- Altruism is present at birth
- Altruism is an exclusively human behaviour
- Performing altruistic behaviours does not improve well-being
- Empathy and altruism are identical

Your answer is incorrect.

EXPLANATION: Altruistic behaviours occur among many animals, including humans. It is well-documented that performing altruistic behaviours has a positive effect on psychological well-being and health. There is also evidence that performing altruistic behaviours relieves physical pain.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 1041.Wang Y et al. Altruistic behaviors relieve physical pain. Proceedings of the National Academy of Sciences. 2020 Jan 14;117(2):950-8.

The correct answer is: Altruism can reduce the physical perception of pain

Question 45

Not answered

Marked out of 1.00

Increased maternal sensitivity during attachment behaviour is due to increased functional connectivity of which structures?

Select one:

- Amygdala and cerebellum
- Hippocampus and amygdala
- Hippocampus and putamen
- Hippocampus and cerebellum
- Putamen and cerebellum

Your answer is incorrect.

EXPLANATION: Maternal sensitivity consists of the accurate interpretation followed by prompt and appropriate response to the infant's signals. It is a well-documented predictor of attachment. The hippocampus and amygdala are key components of the limbic system that are centrally implicated in socio-emotional functioning. The amygdala plays a salient role in orienting to and processing emotionally significant information. The hippocampus is central to learning and memory. The hippocampus and amygdala are activated during interactions between infants and their caregivers.

Ref: Bernier A et al. Mother–infant interaction and child brain morphology: A multidimensional approach to maternal sensitivity. Infancy. 2019 Mar;24(2):120-38.

The correct answer is: Hippocampus and amygdala

Question 46

Not answered

Marked out of 1.00

According to Piaget's theory of development, what is the most important feature of the sensorimotor stage?

Select one:

- Syllogistic reasoning
- Conservation
- Egocentrism
- Object permanence
- Phenomenalistic causality

Your answer is incorrect.

EXPLANATION: According to Piaget, the critical achievement of the sensorimotor stage is the development of object permanence or the schema of the permanent object. This relates to the child's ability to understand that objects have an existence independent of their involvement. Infants learn to differentiate themselves from the world and maintain a mental image of an object, even when it is not visible.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 1007.

The correct answer is: Object permanence

Question 47

Not answered

Marked out of 1.00

If a child wants to fit in socially and be described as a ‘nice person’, he is exhibiting which level of Kohlberg’s theory of morality?

Select one:

- Social order
- Conventional
- Post-conventional
- Pre-conventional
- Self-interest

Your answer is incorrect.

EXPLANATION: Lawrence Kohlberg integrated Jean Piaget’s concepts and described multiple stages of moral development within three significant levels of morality. The first level is pre-conventional morality, in which punishment and obedience are the determining factors. The second level is conventional morality, in which right and wrong are defined in terms of convention and ‘what people would say’. Children try to conform to gain approval and maintain good relationships with others and are also guided by rigid codes of ‘law and order’ (authority orientation). The third and highest level is the morality of self-accepted moral principles (post-conventional), in which children voluntarily comply with rules based on a concept of universal ethical principles and make exceptions to rules in certain circumstances.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 902.

Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015. p. 167.

The correct answer is: Conventional

Question 48

Not answered

Marked out of 1.00

Which of the following is defined as the ‘meaning of words’ in linguistics?

Select one:

- Phonology
- Phenomenology
- Lexical semantics
- Orthography
- Syntax

Your answer is incorrect.

EXPLANATION: Lexical semantics is the study of word meaning. Phonology, orthography, syntax, and phenomenology are all relevant aspects of linguistics, but they deal with different aspects of language, such as sound patterns (phonology), written symbols (orthography), sentence structure (syntax), and subjective experiences (phenomenology), respectively. Lexical semantics specifically deals with the meaning of words themselves.

Ref: Alonso-Sánchez MF, Ford SD, MacKinley M, Silva A, Limongi R, Palaniyappan L. Progressive changes in descriptive discourse in first episode schizophrenia: a longitudinal computational semantics study. Schizophrenia. 2022 Apr 12;8(1):36.

The correct answer is: Lexical semantics

Question 49

Not answered

Marked out of 1.00

Which of the following antidepressants has the least sexual side effects?

Select one:

- Clomipramine
- Amitriptyline
- Sertraline
- Paroxetine
- Vortioxetine

Your answer is incorrect.

EXPLANATION: Of the antidepressants, agomelatine, bupropion, and vortioxetine have the best evidence supporting a more favourable sexual side effect profile. Among the tricyclic antidepressants, sexual dysfunction is more common with clomipramine, amitriptyline, and imipramine. Overall evidence suggests relatively high rates of sexual dysfunction across all the SSRIs.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 405-406.

The correct answer is: Vortioxetine

Question 50

Not answered

Marked out of 1.00

Clozapine has the most affinity to which receptor?

Select one:

- Alpha 1A/B
- 5HT2A
- Alpha 2B/C
- D2/D3
- D4

Your answer is incorrect.

EXPLANATION: Clozapine binds most potently to alpha 1A/B receptors. Though it has 5HT2A/D2 antagonism, numerous binding properties have been identified for clozapine, most of which are more potent than its binding at the D2 receptor. Clozapine's blocking of alpha 1 adrenergic receptors can cause dizziness, hypotension, and syncope.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 217.

Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 193.

The correct answer is: Alpha 1A/B

Question 51

Not answered

Marked out of 1.00

Which of the following drugs is most likely to be abused by students who are studying for exams?

Select one:

- Amphetamine
- Guanfacine
- Clonidine
- Atomoxetine
- Propranolol

Your answer is incorrect.

EXPLANATION: Smart drugs' (also known as 'nootropics') are used by healthy subjects (i.e. exam-taking students and worker) typically to improve memory, attention, learning, executive functions and vigilance. The most popular drugs are amphetamine-type stimulants, methylphenidate, donepezil, selegiline, modafinil, piracetam, benzodiazepine inverse agonists, and unifiram analogues.

Ref: Schifano F et al. Benefits and harms of 'smart drugs'(nootropics) in healthy individuals. Drugs. 2022 Apr;82(6):633-47.

The correct answer is: Amphetamine

Question 52

Not answered

Marked out of 1.00

What is atomoxetine's mechanism of action?

Select one:

- SARI
- a2 agonist
- D2 antagonist
- SSRI
- Selective NRI

Your answer is incorrect.

EXPLANATION: Atomoxetine is a selective norepinephrine reuptake inhibitor (NRI). Sometimes called NET inhibitors, the selective NRIs have known antidepressant properties. However, despite its name, atomoxetine enhances both dopamine and norepinephrine in the frontal cortex, presumably accounting for its therapeutic actions on attention and concentration in ADHD.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 407.

Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 83.

The correct answer is: Selective NRI

Question 53

Not answered

Marked out of 1.00

Which type of glutamate receptor is linked to G-proteins?

Select one:

- AMPA
- NMDA
- Metabotropic
- EAAT
- Kainate

Your answer is incorrect.

EXPLANATION: There are several types of glutamate receptors, including the neuronal presynaptic reuptake pump (EAAT) and the vesicular transporter for glutamate into synaptic vesicles (vGluT). Metabotropic glutamate receptors are those glutamate receptors that are linked to G proteins. NMDA, AMPA, and kainate receptors for glutamate are all members of the ligand-gated ion-channel family of receptors (also known as ionotropic receptors or ion-channel-linked receptors).

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 99-101.

The correct answer is: Metabotropic

Question 54

Not answered

Marked out of 1.00

Which of the following antipsychotics reaches peak plasma concentration the earliest?

Select one:

- Aripiprazole
- Sulpiride
- Iloperidone
- Quetiapine
- Olanzapine

Your answer is incorrect.

EXPLANATION: Peak plasma concentrations of: quetiapine are reached in 1-1.5 hours; olanzapine within six hours; sulpiride in 3-6 hours; aripiprazole within 3-5 hours; iloperidone within 2-4 hours.

Ref: Mauri MC et al. Clinical pharmacology of atypical antipsychotics: an update. EXCLI journal. 2014;13:1163.

The correct answer is: Quetiapine

Question 55

Not answered

Marked out of 1.00

In pharmacokinetics, what does ‘area under the curve’ denote?

Select one:

- Time to reduce plasma concentration by ½
- Maximum concentration of the drug
- Time to maximum concentration of the drug
- Time until the drug is fully eliminated
- Total drug exposure over time

Your answer is incorrect.

EXPLANATION: The ‘area under the curve’ (AUC) is the total drug dose from time 0 to infinity. It represents the total drug dose or drug burden.

Ref: Currie GM. Pharmacology, part 2: introduction to pharmacokinetics. Journal of Nuclear Medicine Technology. 2018 Sep 1;46(3):221-30.

The correct answer is: Total drug exposure over time

Question 56

Not answered

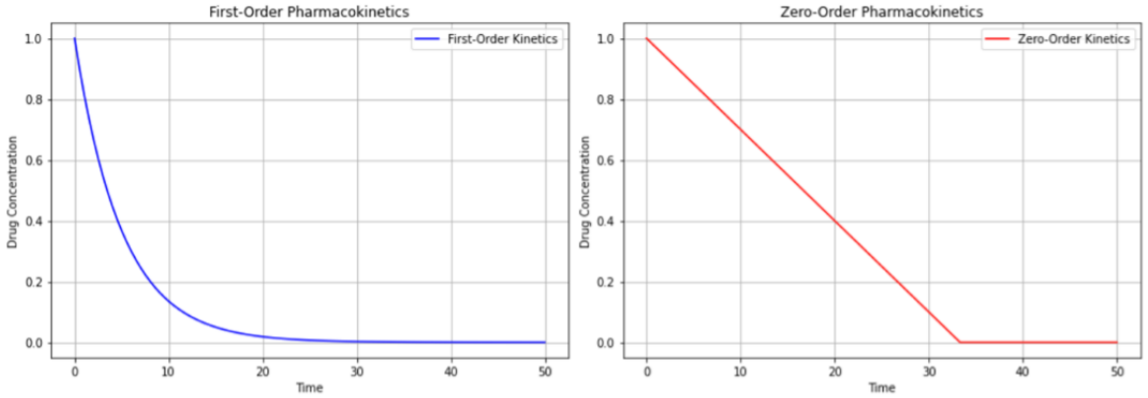
Marked out of 1.00

Which of the following is true in first order kinetics?

Select one:

- A constant fraction of drug is eliminated per unit of time
- A constant amount of drug is eliminated per unit of time
- A log of plasma concentration plotted against time will show non-linear decay
- A graph of plasma concentration plotted against time will show linear decay
- The rate of elimination is independent of drug concentration

Your answer is incorrect.



First-order kinetics (blue plot) is when a constant (exponential) fraction of drug is eliminated per unit of time. It is similar to the concept of radioactive decay. First-order elimination follows an exponential trend and can be displayed with a logarithmic y-axis to generate a straight line. Zero-order kinetics (red plot) is when there is a constant (linear) rate of drug elimination, indicating that the rate of elimination is independent of drug concentration. Unlike first-order kinetics, a constant amount of drug is eliminated per unit of time.

Plot created using jupyter notebook (matplotlib)

Ref: Currie GM. Pharmacology, part 2: introduction to pharmacokinetics. Journal of Nuclear Medicine Technology. 2018 Sep 1;46(3):221-30.

The correct answer is: A constant fraction of drug is eliminated per unit of time

Question 57

Not answered

Marked out of 1.00

Absorption of a drug is generally independent of

Select one:

- Site of absorption
- Elimination of the drug
- Form of the drug
- Chemical nature of the drug
- Route of administration

Your answer is incorrect.

EXPLANATION: Factors that influence drug absorption include the molecular size of the drug, lipid solubility of the drug, degree of ionisation of the drug, dosage form (e.g., tablet or solution), chemical nature of the drug, whether there is complex formation, pharmacological effect of the dose and concentration of the drug, blood flow (at the side of administration), site of absorption, and route of administration.

Ref: Currie GM. Pharmacology, part 2: introduction to pharmacokinetics. Journal of Nuclear Medicine Technology. 2018 Sep 1;46(3):221-30.

The correct answer is: Elimination of the drug

Question 58

Not answered

Marked out of 1.00

When patients comply with medications simply because the drugs are recommended by doctors, this is best attributed to which type of power, according to the classic work of French & Raven?

Select one:

- Coercive
- Expert
- Referent
- Legitimate
- Informational

Your answer is incorrect.

EXPLANATION: According to French & Raven, expert power stems from the target (patient) attributing superior knowledge and/or ability to the influencing agent (doctor). Explanations or reasons are not necessary for the patient to comply with the doctor's recommendations.

Ref: Brown JH, Raven BH. Power and Compliance in Doctor/Patients Relationship. REVISTA LEX MERCATORIA Doctrina, Praxis, Jurisprudencia y Legislación. 1994;6(1):3-22.

The correct answer is: Expert

Question 59

Not answered

Marked out of 1.00

Ebstein anomaly is most associated with which drug?

Select one:

- Clonazepam
- Lamotrigine
- Lithium
- Valproate
- Carbamazepine

Your answer is incorrect.

EXPLANATION: Lithium use during pregnancy has a well-known association with the cardiac malformation, Ebstein anomaly. Of note, however, is that more recent data suggest that the magnitude of the effect is much smaller than previously estimated.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 690.

The correct answer is: Lithium

Question 60

Not answered

Marked out of 1.00

Metabolic syndrome was present in what percentage of patients in the CATIE trial?

Select one:

- 41-50%
- 31-40%
- 61-70%
- 11-20%
- 81-90%

Your answer is incorrect.

EXPLANATION: In the CATIE trial, metabolic syndrome prevalence was 41% and 43%, respectively, using the National Cholesterol Education Program (NCEP) and AHA (fasting glucose threshold of 100mg/dl) derived criteria.

Ref: McEvoy JP et al. Prevalence of the metabolic syndrome in patients with schizophrenia: baseline results from the Clinical Antipsychotic Trials of Intervention Effectiveness (CATIE) schizophrenia trial and comparison with national estimates from NHANES III. Schizophrenia research. 2005 Dec 1;80(1):19-32.

The correct answer is: 41-50%

Question 61

Not answered

Marked out of 1.00

Hallucinogenic effects of psychedelics are produced by activity on which receptor?

Select one:

- 5HT2
- HT1
- 5HT3
- 5HT1
- 5HT7

Your answer is incorrect.

EXPLANATION: Hallucinogens such as LSD, psilocybin, and mescaline are 5HT2A agonists. When these agents stimulate 5HT2A receptors on glutamatergic pyramidal neurons in the prefrontal cortex, this causes overactivation of the glutamate neuron. The resultant release of glutamate into the ventral tegmental area causes hyperactivity of the mesolimbic dopamine pathway, resulting in delusions and auditory hallucinations. Excessive glutamate release in the visual cortex can cause visual hallucinations.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 135.

The correct answer is: 5HT2

Question 62

Not answered

Marked out of 1.00

Which of the following drugs has the narrowest therapeutic index?

Select one:

- Carbamazepine
- Doxepin
- Clomethiazole
- Zopiclone
- Clonazepam

Your answer is incorrect.

EXPLANATION: Some drugs have a narrow therapeutic index (a small increase in dose can cause toxicity and a small reduction in dose can cause a loss of therapeutic action). The most commonly prescribed ones are digoxin, warfarin, theophylline, phenytoin, carbamazepine, and lithium.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 602, 787.

The correct answer is: Carbamazepine

Question 63

Not answered

Marked out of 1.00

Which of the following is the most biologically important metabolite?

Select one:

- Lorazepam
- Alprazolam
- Chlordiazepoxide
- Oxazepam
- Diazepam

Your answer is incorrect.

EXPLANATION: Oxazepam is a short-acting benzodiazepine and one of the most frequently prescribed anxiolytic drugs. It is also the active metabolite of a wide range of other benzodiazepines, such as diazepam, ketazolam, temazepam, and chlordiazepoxide.

Ref: Dinis-Oliveira RJ. Metabolic profile of oxazepam and related benzodiazepines: clinical and forensic aspects. Drug metabolism reviews. 2017 Oct 2;49(4):451-63.

The correct answer is: Oxazepam

Question 64

Not answered

Marked out of 1.00

Which of the following best describes the primary mechanisms of action of lurasidone?

Select one:

- Antagonist at D2, 5HT2A, and 5HT7
- Antagonist at D2; agonist at 5HT2A and 5HT7
- Agonist at D2; antagonist at 5HT2A and 5HT7
- Agonist at D2, 5HT2A, and 5HT7
- Antagonist at H1 and M1

Your answer is incorrect.

EXPLANATION: Lurasidone is a 5HT2A/D2 antagonist. It also potently blocks 5HT7 receptors, which may be beneficial for mood, sleep, cognitive impairment, and negative symptoms in schizophrenia, and also in bipolar disorder and major depressive disorder. Lurasidone lacks potent actions at D1, M1, and H1 receptors, theoretically suggesting less propensity for inducing cognitive impairment, weight gain, or sedation compared to other agents with these properties.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 236.

Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 457.

The correct answer is: Antagonist at D2, 5HT2A, and 5HT7

Question 65

Not answered

Marked out of 3.00

Disorders of Visual Perception

Choose the term that best applies to each of following scenarios.

1. When a man with schizophrenia first looks at a friend's face, it appears normal, but the face then transforms to be frighteningly grotesque and misshapen.

2. Despite not having any pets, an elderly woman with visual impairment intermittently sees cats in her apartment, which delight her.

3. Upon seeing a shadow, a recently bereaved man briefly thinks that it is his deceased wife.

CORRECT ANSWERS:

1) When a man with schizophrenia first looks at a face, it appears normal, but the face then transforms to be frighteningly grotesque and misshapen. - Paraprosopia

2) Despite not having any pets, an elderly woman with visual impairment intermittently sees small cats in her apartment, which delight her. – Hallucination

3) Upon seeing a shadow, a recently bereaved man briefly thinks that it is his deceased wife. - Illusion

EXPLANATION: Metamorphopsia or dysmegalopsia refers to change in the perceived shape of an object. When metamorphopsia affects faces, it is referred to as paraprosopia. Faces generally transform into a grotesque mask (monster, vampire, werewolf). Autoparaprosopia is a term used to describe a phenomenon where an individual perceives a transformation or alteration in their own facial appearance while gazing into a mirror. Note that paraprosopia is a distortion of reality seen in conditions such as schizophrenia and is seen as a delusional misidentification similar to Capgras and Fregoli delusions), while the term prosopagnosia is an inability to identify faces seen in organic conditions affecting brain's face recognition area.

Charles Bonnet syndrome is a condition in which complex visual hallucinations occur in the absence of any psychopathology and in clear consciousness. It is associated with either central or peripheral reduction in vision and not surprisingly is most common in the elderly, although it can occur in younger people also. These visual hallucinations can be accompanied by pleasure and amusement.

In illusions, stimuli from a perceived object are combined with a mental image to produce a false perception. Affect illusions arise in the context of a particular mood state. For example, a bereaved person may momentarily believe they 'see' the deceased person.

Ref: Casey P, Kelly B. Fish's clinical pathology: Signs and symptoms in psychiatry, 4th Edition. 2019. pp. 25-26, 29, 32.

Oyebode F. Sims' symptoms in the mind: Textbook of descriptive psychopathology. Elsevier Health Sciences; 2022 (page 90).

Martini F, Fregna L, Vanelli I, Bigai G, Tonet L, Cavallaro R. Psychopathology. In Fundamentals of Psychiatry for Health Care Professionals 2022 Aug 31 (pp. 15-47). Cham: Springer International Publishing.

Question 66

Not answered

Marked out of 5.00

Tests of neuropsychological functioning

Choose the test that best assesses the following neuropsychological functions.

1. Motor speed ✖
2. Abstract reasoning ✖
3. Working memory ✖
4. Visuospatial construction ✖
5. Set shifting ✖

CORRECT ANSWERS:

1) Motor speed – Finger Tapping

2) Abstract reasoning – Interpreting Proverbs

3) Working memory – Digit Span

4) Visuospatial construction – Clock Drawing

5) Set shifting – Wisconsin Card Sorting

EXPLANATION: Finger tapping is a standard measure of simple motor speed. It is particularly useful for documenting lateralised motor impairment.

Abstract reasoning is the ability to shift back and forth between general concepts and specific examples. Having the patient identify similarities between like objects or concepts (apple and pear, bus and airplane, or a poem and a painting) as well as interpreting proverbs are useful ways to assess one's ability to abstract.

Working memory is synonymous with short-term memory, which is responsible for the immediate recall of small amounts of verbal (as in digit span) or visuospatial information.

Clock drawing is a useful technique for assessing constructional ability and is also sensitive to organisation and planning.

The best formal test of set shifting is the Wisconsin Card Sorting Test, which also involves problem-solving and hypothesis testing.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 11, 53.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 82.

Hodges JR. Cognitive assessment for clinicians, 3rd Edition. 2018. p. 135.

Question 67

Not answered

Marked out of 5.00

Neurons and Glial Cells

Identify the cell responsible for each of the following functions.

1. Phagocytosis of cellular debris ✖
2. Nutrition of neurons and release of glutamate ✖
3. GABA-ergic projection in the striatum ✖
4. Myelination in the central nervous system ✖
5. Myelination in the peripheral nervous system ✖

CORRECT ANSWERS:

1) Phagocytosis of cellular debris - Microglia
2) Nutrition of neurons and release of glutamate - Astrocytes
3) GABA-ergic projection in the striatum – Medium spiny neurons
4) Myelination in the central nervous system – Oligodendrocytes
5) Myelination in the peripheral nervous system – Schwann cells

EXPLANATION: There are three types of glial cells. The most common are the astrocytes, which have several functions, including nutrition of neurons, deactivation of some neurotransmitters, and integration with the blood-brain barrier. The astrocyte carries most of the extracellular glutamate. The oligodendrocytes in the central nervous system and the Schwann cells in the peripheral nervous system wrap their processed around neuronal axons, resulting in myelin sheaths that facilitate the conduction of electrical signals. The third type of glial cells, the microglia, which are derived from macrophages, are involved in removing cellular debris following neuronal death. The functions of microglia in the non-diseased brain probably include a role in synaptic maintenance.

GABA-ergic medium spiny neurons (MSNs), the principal neuron type in the striatum, are projection neurons with predominantly dopaminergic and glutamatergic inputs. The loss of MSNs in the striatum is the hallmark of Huntington disease.

Ref: Siracusa R et al. Astrocytes: role and functions in brain pathologies. Frontiers in pharmacology. 2019 Sep 27;10:1114.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 924.

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Question 68

Not answered

Marked out of 5.00

Specialised Cells

Identify the cell responsible for each of the following functions. (Answers may be used more than once.)

1. Cells responsible for rapid signal transmission between brain regions

2. Principal neuron type in the cerebral cortex responsible for cortical processing

3. Specialized neurons associated with social cognition and self-awareness.

4. Inhibitory interneurons that regulate neural activity in cortical circuits

5. Star-shaped glial cells involved in synaptic integration in the brain.

CORRECT ANSWERS:

1. Cells responsible for rapid signal transmission between distant brain regions - Pyramidal cells
2. Principal neuron type in the cerebral cortex responsible for cortical processing - Pyramidal cells
3. Specialized neurons associated with social cognition and self-awareness - Von Economo neurons
4. Inhibitory interneurons that regulate neural activity in cortical circuits - Basket cells
5. Star-shaped glial cells involved in synaptic integration in the brain - Astrocytes

EXPLANATION: A diverse set of specialised cell types collectively contribute to the complex functions of the brain. Pyramidal cells serve as the principal excitatory neurons responsible for rapid signal transmission between brain regions and cortical processing in the cerebral cortex. von Economo in 1925 did pioneering work on the cytoarchitectonics of the cerebral cortex and discovered a specialized and unique cell

type in the fronto-insular and anterior cingulate cortex. These von Economo neurons (VENs) appear as large sticks, rods or corkscrew cells because of their extremely elongated and relatively thin cell body which is distinct from the more common oval or spindle-shaped infragranular neurons. Basket cells, specialised cells first described by Ramon y Cajal, are inhibitory interneurons in the neocortex, cerebellum and hippocampus; they form 'baskets' of axonal arborizations onto and around the somata of target neurons. Astrocytes, characterized by their star-shaped morphology, play a multifunctional role in the brain, including synaptic integration, regulation of neurotransmitters, nutrient support for neurons and overall brain homeostasis.

Ref: Siracusa R et al. Astrocytes: role and functions in brain pathologies. Frontiers in pharmacology. 2019 Sep 27;10:1114. Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 924.

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Banovac I, Sedmak D, Judas M, Petanjek Z. Von Economo neurons—primate-specific or commonplace in the mammalian brain?. Frontiers in Neural Circuits. 2021 Sep 1;15:714611.

Question 69

Not answered

Marked out of 3.00

Adverse effects of psychotropic medication

Identify the drug most likely to have caused the following side effects.

1. Weight loss ✖

2. Worsening psoriasis ✖

3. Rash ✖

CORRECT ANSWERS:

1) Weight loss - Topiramate
2) Worsening psoriasis – Lithium
3) Rash - Lamotrigine

EXPLANATION: Topiramate can cause nausea, appetite loss, and weight loss. Topiramate is an antiepileptic medication that is also used as an adjunct treatment for mood disorders and binge-eating disorder. One of its notable side effects is appetite suppression, which often leads to weight loss in individuals taking the medication. Patients may also experience nausea, which can contribute to decreased food intake and, consequently, weight loss.

Some skin conditions such as psoriasis and acne can be aggravated by lithium.

10-14% of patients receiving lamotrigine will develop rash. Most are benign. A minority may be serious/life-threatening skin reactions requiring hospitalisation (e.g., Steven-Johnson syndrome, toxic epidermal necrolysis, angio-oedema, and a rash associated with several systemic manifestations).

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. pp. 781-782.

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